



SRK INSTITUTE OF MANAGEMENT AND COMPUTER EDUCATION

Affiliated to



Kranti guru Shyamji Krishna Verma University

A Project Report on

Expense Tracker Android Application

Submitted in partial fulfilment of the requirements for the Mobile Application Development, as
a part of

BACHELOR OF COMPUTER APPLICATIONS (BCA)

Submitted by:

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Acknowledgment

I take this opportunity to express my heartfelt gratitude to all those who have helped me directly or indirectly in the successful completion of this project, “**Expense Tracker Android Application.**”

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Abstract

In the modern era, financial management has become a vital part of every individual's daily life. With the rapid increase in expenses and multiple spending categories, keeping track of every transaction manually has become a challenge. To address this issue, the **Expense Tracker Android Application** has been developed to provide a convenient and efficient solution for managing personal finances. This mobile-based application allows users to record their income and expenses, analyze their spending patterns, and maintain better control over their financial habits. It serves as a personal assistant that helps users to stay organized, plan budgets, and make informed decisions about their expenditures.

The application provides various features such as user authentication through login and registration, the ability to add, edit, or delete expenses, view categorized expense summaries, and visualize financial data through graphical charts. One of its key features is the budget management system that alerts users when they exceed their preset limits, helping them to stay financially disciplined. The app also includes options to export data in CSV format, ensuring that users can back up and review their spending history easily. The interface is designed to be simple and user-friendly, ensuring smooth navigation and a pleasant user experience for individuals of all age groups.

Technically, the system has been implemented using **Kotlin** as the programming language within **Android Studio**, while **XML** has been used for front-end layout design to ensure a responsive and visually appealing interface. **SQLite**, a lightweight local database, has been integrated for secure and efficient data storage, allowing the application to function offline without requiring an internet connection. This project not only showcases the development of a practical and reliable financial tool but also provides hands-on experience in Android app development, UI/UX design, and database integration. The **Expense Tracker Android Application** thus serves as a complete example of how mobile technology can be leveraged to simplify day-to-day financial management and enhance user productivity.

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Introduction

In today's digital world, **mobile applications** have become an integral part of human life, simplifying almost every activity—from communication and education to entertainment, health, and financial management. With the continuous advancement in smartphone technology and the availability of affordable internet access, mobile apps now play a crucial role in providing convenient, real-time solutions to users' everyday needs. One of the most practical and widely used categories of mobile applications is **personal finance management**, which allows individuals to manage their income, expenses, and savings effortlessly through their mobile devices. Mobile apps are no longer limited to luxury or entertainment—they have become essential tools for organization, productivity, and financial control.

The **Expense Tracker Android Application** has been developed with this objective in mind—to help individuals efficiently manage their daily, weekly, and monthly financial transactions. Managing expenses manually is often time-consuming and error-prone. Many people face difficulties keeping track of where their money is being spent and how to maintain a balance between income and expenditure. This application provides a **simple, lightweight, and user-friendly platform** that enables users to record, categorize, and monitor their expenses easily. By offering features such as budget limit setup, automatic alerts when the limit is exceeded, and visual representation of spending through charts, the app empowers users to make informed financial decisions and develop better money management habits.

The **scope of this project** includes developing a fully functional Android-based application that performs expense tracking, budget management, and data analysis without requiring an internet connection. It integrates essential modules such as user login and registration, expense addition and modification, chart visualization, CSV data export, and budget alerts. Designed using Kotlin and XML within Android Studio, the app leverages SQLite as its local database to store user data securely. The project not only focuses on delivering an effective personal finance management solution but also demonstrates key aspects of Android development, database integration, and user interface design. Ultimately, the **Expense Tracker Android Application** serves as a complete, practical tool for users who wish to gain better control over their financial activities through a modern, mobile-based approach.

Objectives of the Project

The main objective of this project is to design and develop a **mobile-based Expense Tracker Application** that enables users to efficiently manage and monitor their financial activities. In today's fast-paced lifestyle, people often struggle to keep track of their daily expenditures and overall spending habits. Hence, the project aims to create a simple, reliable, and user-friendly solution that assists individuals in maintaining their financial discipline and achieving better control over their money. This application not only simplifies expense tracking but also helps users analyze their spending behavior, plan budgets, and make more informed financial decisions.

In addition to developing the application, this project was also undertaken as a practical learning experience to strengthen the understanding of **mobile application development concepts** using **Android Studio** and **Kotlin** programming. Through this project, several technical and conceptual goals were established and achieved. The major objectives of the project are summarized as follows:

1. **To develop an Android-based Expense Tracker Application** that allows users to record, categorize, and manage their daily, weekly, and monthly expenses with ease.
2. **To design a user-friendly interface** using XML layouts and modern UI/UX design principles that ensure smooth navigation and an engaging user experience.
3. **To implement secure and efficient local data storage** using **SQLite database**, enabling the app to function fully offline without relying on cloud-based systems like Firebase.
4. **To provide features for setting and managing budgets**, including notifications or alerts when users exceed their predefined spending limits.
5. **To generate visual reports and charts** that help users understand their spending patterns and financial habits more clearly.
6. **To include functionalities for exporting expense data** in CSV format, allowing users to back up or analyze their data externally.
7. **To enhance programming and problem-solving skills** in Kotlin by applying real-world logic to develop a complete and functioning Android application.
8. **To integrate good software engineering practices**, such as modular coding, debugging, and version control, to improve the efficiency and reliability of the project.
9. **To gain hands-on experience in Android app lifecycle management, event handling, and UI binding** through practical implementation within Android Studio.
10. **To develop a scalable foundation** for future enhancements such as AI-based budget suggestions, cloud synchronization, and advanced data analytics.

In conclusion, the objective of this project is not only to produce a fully operational expense tracking system but also to serve as a comprehensive learning platform that bridges theoretical knowledge and practical application in the field of mobile app development. The successful completion of this project has significantly improved the technical expertise, creativity, and analytical skills required to develop professional Android applications.

System Analysis

Existing System

In the traditional method of managing personal expenses, most individuals rely on **manual approaches** such as maintaining handwritten records, spreadsheets, or basic note-taking applications. While these methods are simple, they are also **time-consuming, error-prone, and difficult to maintain consistently**. In many cases, users forget to log their expenses regularly or fail to analyze their overall spending pattern. Manual records also make it challenging to track small transactions, categorize expenses properly, or set financial limits for specific time periods.

Some people may use **generic budgeting tools or basic calculator apps**, but these often lack personalization and automatic data handling. Moreover, many existing digital solutions are web-based or require an internet connection, which is not always available. They may also store sensitive financial data on third-party servers, raising **data privacy and security concerns**. As a result, users need a **simple, secure, and offline-capable** system that can manage financial data efficiently without technical complexity.

Thus, the existing systems fail to provide a comprehensive, secure, and easily accessible solution for users who want to track their daily, weekly, and monthly expenses in an organized way.

Proposed System

The **Expense Tracker Android Application** is proposed to overcome the drawbacks of existing manual and online systems by providing a **lightweight, offline, and user-friendly mobile application**. The proposed system helps users maintain accurate financial records, visualize their spending behavior, and set clear budget goals directly on their smartphones.

This application enables users to **add, update, or delete expense entries**, categorize transactions (such as food, transport, shopping, or utilities), and automatically calculate the total amount spent within a specific period. It allows users to define a **budget limit** and receive an alert whenever the spending exceeds that limit. Furthermore, the app provides **visual insights through charts and graphs**, helping users analyze their financial habits more effectively.

Unlike many online systems, this app uses **SQLite as a local database**, ensuring that user data remains stored safely within the device itself. It operates entirely offline, ensuring **data privacy, reliability, and uninterrupted performance** even without an internet connection. The system also includes a **data export feature** that allows users to download their financial data in CSV format, ensuring easy portability and backup.

Overall, the proposed system offers a **modern, secure, and highly efficient solution** for individual expense management, eliminating the inconvenience and inaccuracy of manual methods.

Feasibility Study

A **feasibility study** helps determine whether the proposed system is practical, realistic, and beneficial from technical, operational, and economic perspectives. The study ensures that the system can be successfully implemented within the available resources and time frame.

-> *Technical Feasibility*

The project is technically feasible because all required tools, technologies, and hardware are easily available. The application is developed using **Android Studio**, which provides an integrated environment for designing, coding, debugging, and testing Android apps. **Kotlin** is used as the primary programming language due to its concise syntax, strong typing, and compatibility with Java-based Android APIs. **XML** has been utilized for user interface design to create responsive and interactive layouts.

For data storage, the app uses **SQLite**, which is embedded directly within the Android framework, eliminating the need for additional database servers or internet connectivity. The hardware and software requirements are minimal, as the app can run smoothly on any Android device with standard specifications. Thus, the system is completely **technically feasible** to develop, deploy, and maintain.

-> *Operational Feasibility*

From an operational standpoint, the system is **highly practical and user-friendly**. The user interface is designed to be simple and intuitive so that even users with limited technical knowledge can easily navigate through the app. Features such as category selection, expense addition, budget alerts, and graphical analysis ensure smooth user interaction and an engaging experience.

Because the app works offline, users do not need to worry about network connectivity or data security. It ensures complete control over personal financial data and eliminates dependency on external servers. The app is also lightweight, requires minimal storage, and loads quickly, making it operationally efficient on a wide range of Android devices. Therefore, the system is **operationally feasible** and suitable for everyday personal use.

-> Economic Feasibility

Economically, the system is **cost-effective** and practical to develop. All tools used in the development—such as Android Studio, Kotlin SDK, and SQLite—are open-source and freely available. The project does not require any paid APIs, hosting services, or cloud subscriptions. Since the app is fully offline and does not rely on third-party servers, there are no recurring maintenance or data storage costs.

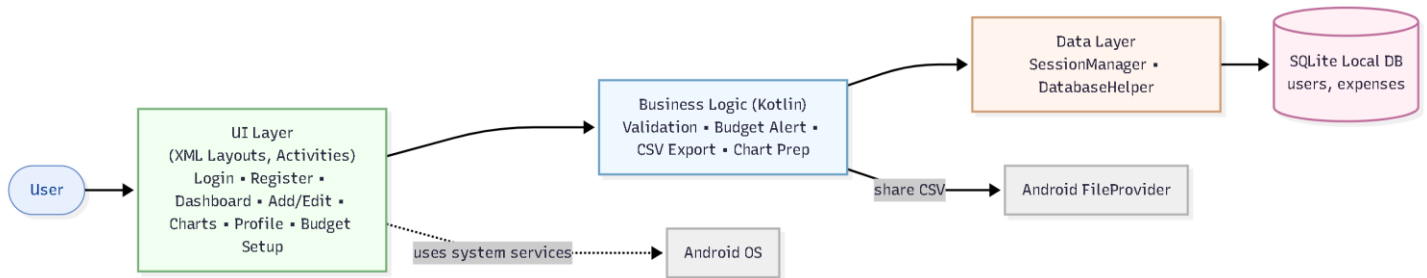
The development and deployment process requires only a basic Android device and a personal computer with the necessary software installed, both of which are easily accessible to students and developers. Therefore, the **overall cost of development is minimal**, making the project highly feasible from an economic standpoint. The app can also be easily upgraded or monetized in the future without significant additional investment.

Conclusion

In summary, the **system analysis** shows that the proposed Expense Tracker Application successfully overcomes the limitations of existing manual and online systems. The app is technically sound, operationally efficient, and economically viable. It provides users with a complete, offline solution for managing their personal finances effectively, ensuring both convenience and data security

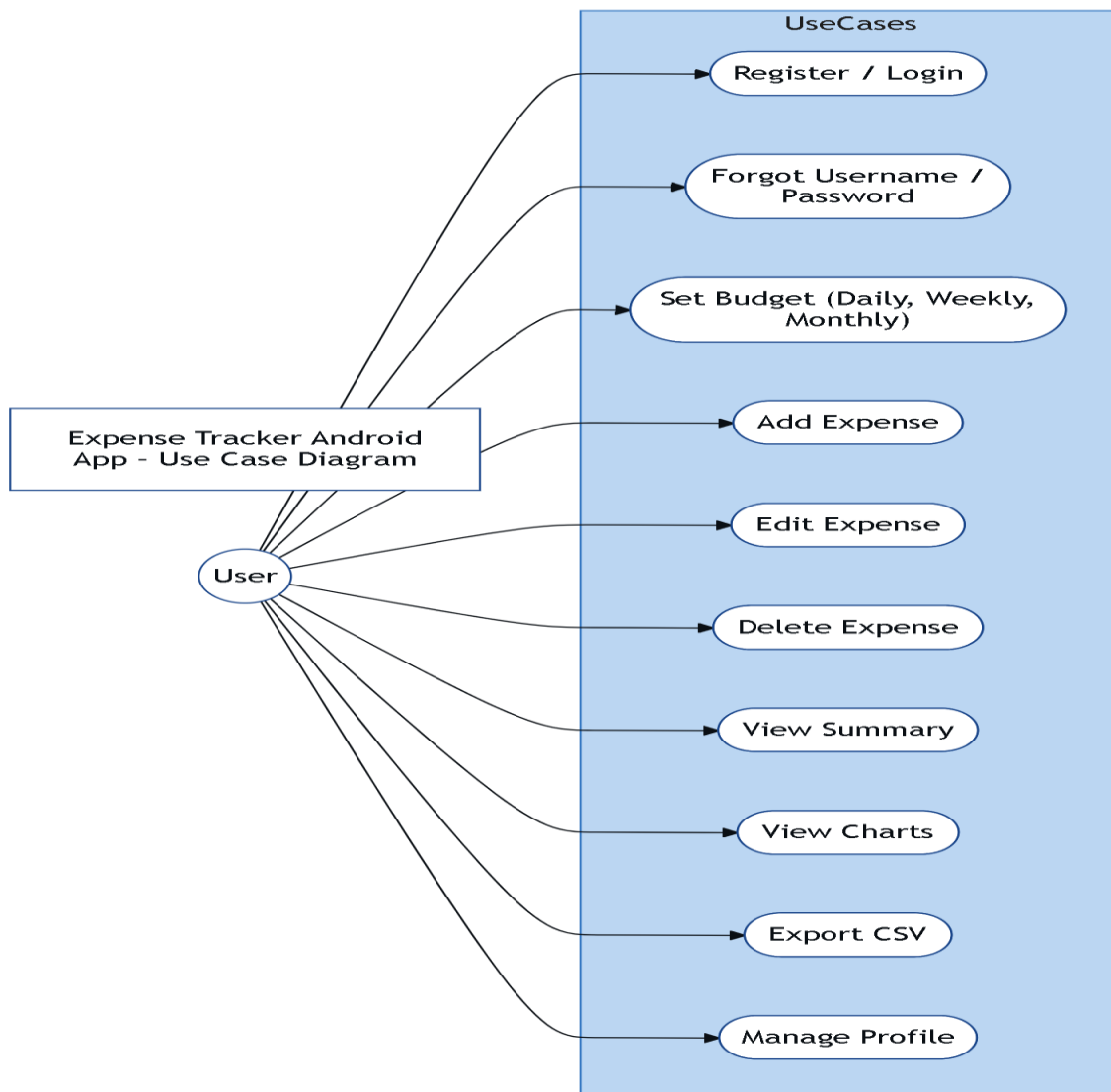
System Design

❖ System Architecture Diagram

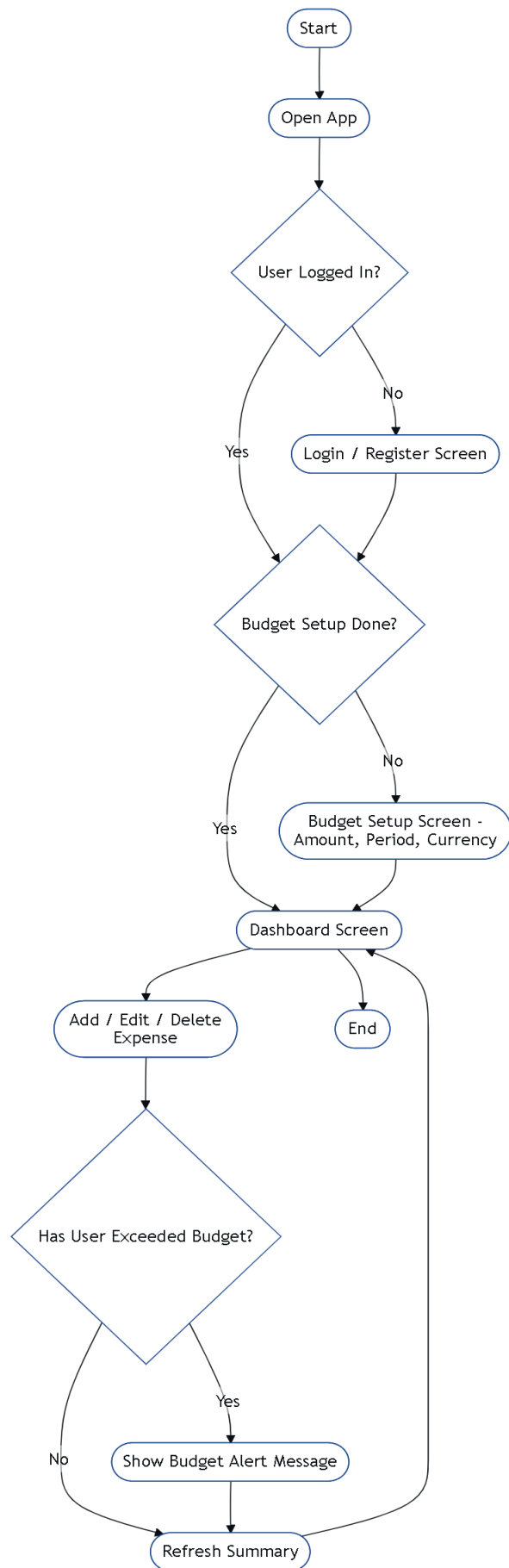


❖ UML Diagrams :

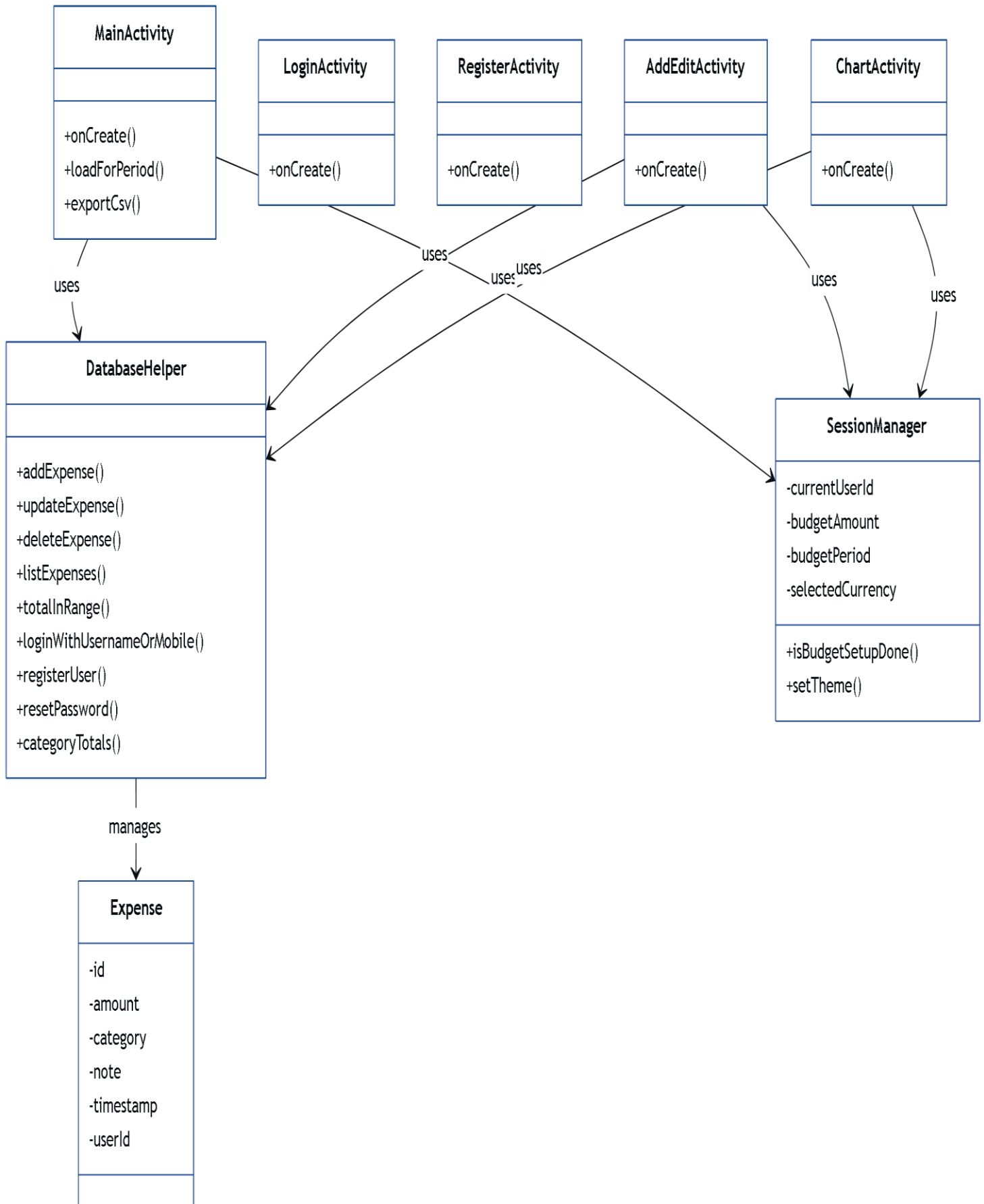
○ Use Case Diagram



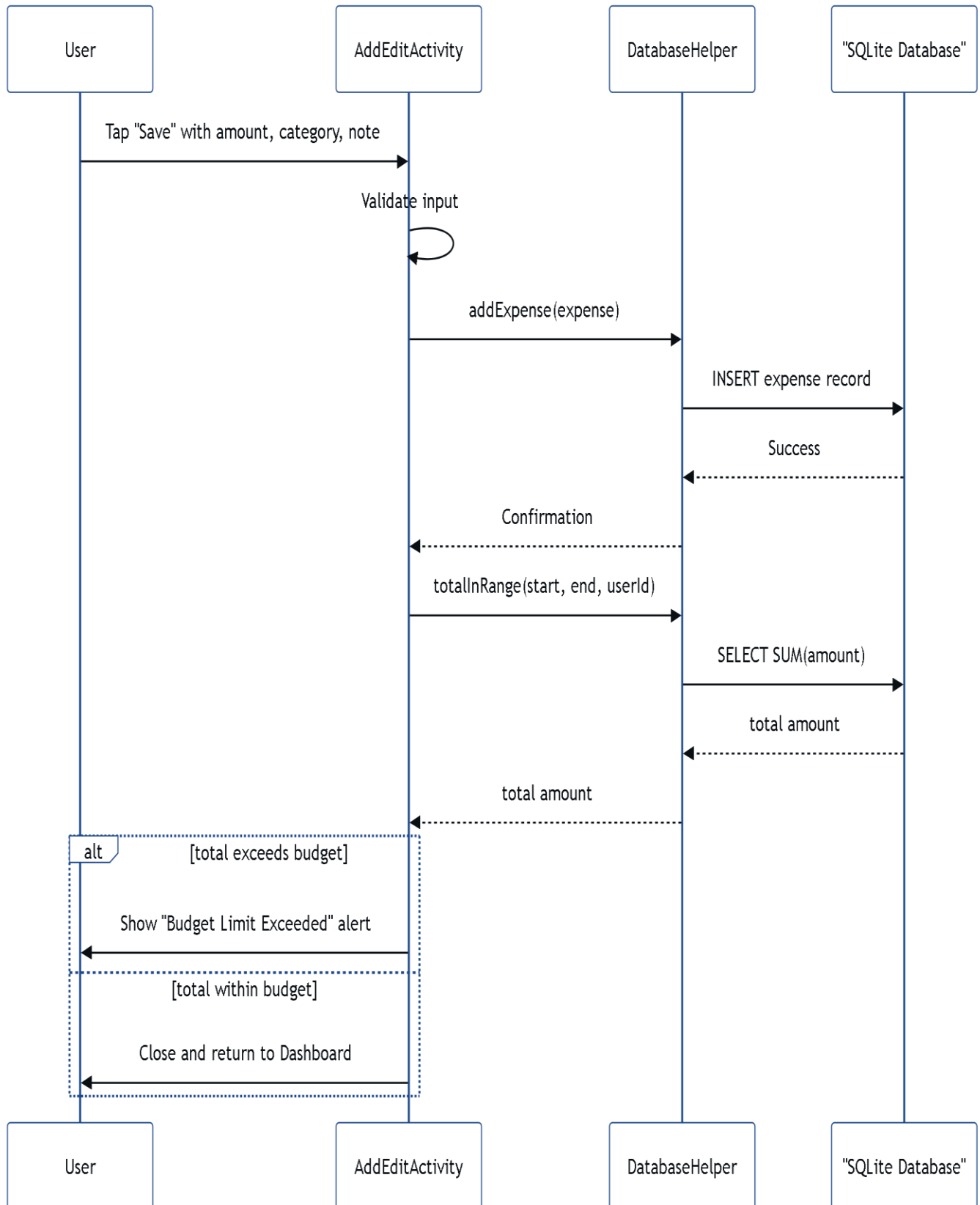
○ Activity Diagram



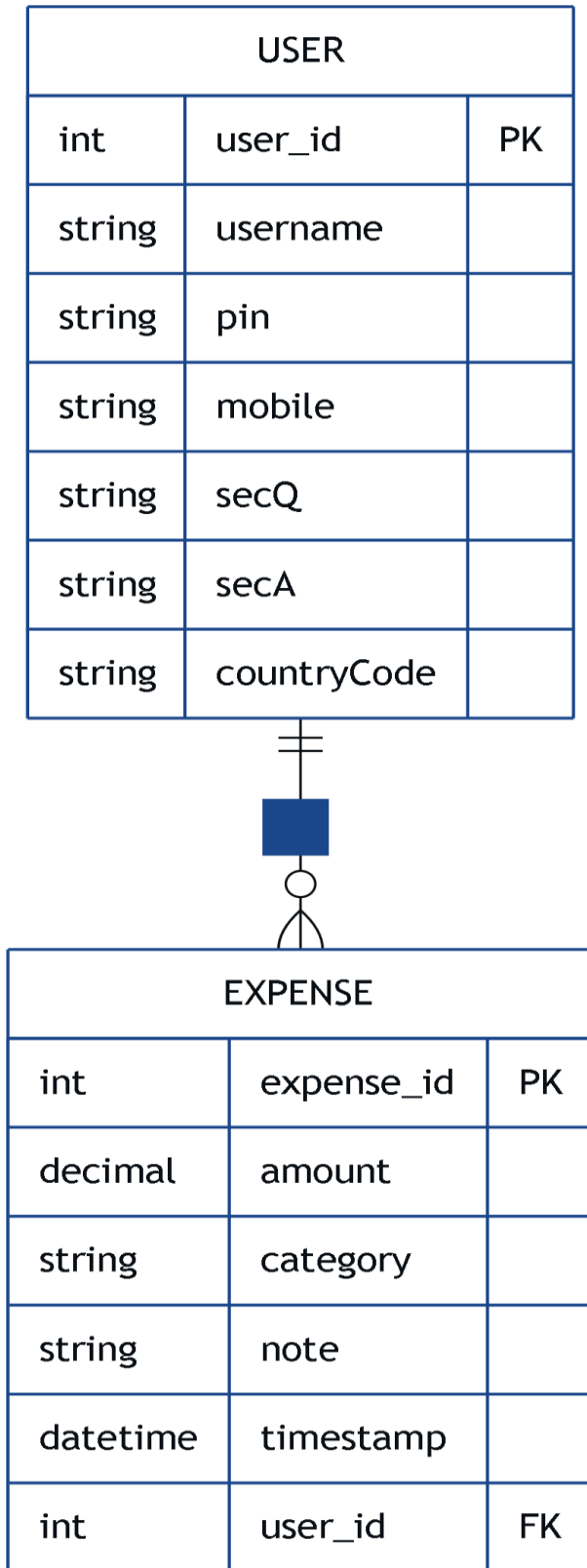
○ **Class Diagram**



○ **Sequence Diagram**



❖ ER Diagram



System Requirements

The successful development and execution of the *Expense Tracker Android Application* require a combination of suitable hardware and software components. The system is designed to function efficiently on modern personal computers for development and standard Android smartphones for end-users. The following are the hardware and software requirements considered for this project.

❖ Hardware Requirements

Component	Specification	Description
Processor	Intel Core i3 / AMD Ryzen 3 or higher	Required for smooth execution of Android Studio and emulators.
RAM	Minimum 8 GB (Recommended 16 GB)	Ensures stable performance while compiling and testing the app.
Storage	At least 10 GB free disk space	Needed for Android Studio, SDK files, Gradle builds, and project files.
Display	1366 × 768 resolution or higher	For better visibility of UI layout and coding interface.
Mobile Device	Android smartphone (Android 8.0 or above) with 3 GB RAM and 100 MB free storage	Used for installing, testing, and running the final application.

Note: The app is lightweight and optimized to run smoothly even on mid-range devices.

❖ Software Requirements

Component	Specification / Version	Description
Operating System	Windows 10 / 11 (64-bit)	Used as the primary development environment.
IDE (Integrated Development Environment)	Android Studio (Version Giraffe or newer)	Provides tools for UI design, coding, testing, and debugging.
Programming Languages	Kotlin, XML	Kotlin is used for backend logic; XML is used for UI design and layout.
Database	SQLite (Local Database)	Used for storing user details, expenses, budgets, and security questions.
Build Tools	Gradle (latest stable version)	Handles compilation, dependency management, and build automation.
Android SDK	Android SDK Platform 33 (Android 13)	Required for compiling and running the Android application.
Emulator / Physical Device	Pixel Emulator (API Level 33) or physical Android phone	Used for testing the application interface and functionality.
Libraries Used	AndroidX Libraries, Material Design Components	Provide UI consistency, backward compatibility, and visual enhancements.

Implementation

This chapter explains **how the Expense Tracker Android Application was developed**, including the tools, technologies, languages, and modular design used. It also demonstrates the practical working of the system through screenshots of the main application screens.

➤ 6.1 Tools and Technologies Used

The following tools and platforms were used during the development of the project:

Tool / Platform	Purpose
Android Studio (Giraffe or higher)	Integrated Development Environment (IDE) for building and testing the Android app.
Kotlin	Primary programming language used for application logic and event handling.
XML (Extensible Markup Language)	Used for designing the app’s user interface layouts.
SQLite	Local database for storing user data, expense records, and settings securely.
Android SDK (API 33)	Provides necessary libraries and components to build Android applications.
Gradle	Build automation tool used to compile, manage dependencies, and generate APK files.
Material Design Components	Used to enhance the user interface and provide a modern visual appearance.
Emulator / Physical Android Device	Used for testing and debugging the application.

➤ 6.2 Programming Languages / Frameworks / APIs

1. **Programming Language:**
 - **Kotlin** was chosen for its modern syntax, safety features, and official support by Google for Android development. It allows for better performance, null-safety handling, and concise code.
2. **UI Framework:**
 - The **Android XML Layout Framework** was used to define screens, buttons, input fields, and overall visual components.

3. Database API:

- The application uses **SQLite** with Android's built-in **SQLiteOpenHelper** class to store and manage data such as user profiles, expense entries, and budget information.

4. Libraries and APIs Used:

- **AndroidX Libraries** for backward compatibility.
- **Material Components** for UI elements like buttons, text fields, and dialogs.
- **FileProvider API** for secure sharing of CSV exports.
- **Pie Chart Custom View / Canvas Drawing API** for displaying visual spending charts.

➤ 6.3 Modules Description

The Expense Tracker Application is divided into several interrelated modules to ensure clarity and modularity:

Module	Description
1. User Authentication Module	Handles user login, registration, password recovery, and session management.
2. Expense Management Module	Allows users to add, edit, and delete expenses with category and note details.
3. Budget Setup Module	Enables users to define their daily, weekly, or monthly budget and track spending limits.
4. Expense Analysis Module	Displays category-wise summaries using a Pie Chart and provides visual insights.
5. Profile & Settings Module	Lets users view profile details, change budget settings, and export expenses to CSV.
6. Alert & Validation Module	Triggers warning popups when the user crosses their budget limit and validates input fields (like 10-digit mobile number).

Testing

This chapter describes the testing strategy and outcomes for the **Expense Tracker Android Application**. A multi-layered approach was followed to ensure correctness, stability, and usability:

- **Unit Testing** (methods, validations, and utilities)
- **Integration Testing** (Activity ↔ DatabaseHelper ↔ SQLite, SessionManager flows)
- **Functional/System Testing** (end-to-end user flows)
- **User Acceptance Testing (UAT)** (real user feedback against requirements)

Test Environment

- Device: Android 11/13 (physical + emulator)
- IDE: Android Studio (Giraffe+)
- Language/Stack: Kotlin, XML, SQLite
- Build: Gradle (debug build)
- Test Data: Local SQLite with dummy users and sample expenses

Unit Testing

Unit tests validated **individual functions** and **business rules**:

- Input validation (e.g., PIN numeric and length, mobile = 10 digits)
- SessionManager getters/setters (budget amount/period/currency, setup flag)
- DatabaseHelper methods (CRUD for expenses, totalInRange, categoryTotals)
- Budget alert trigger logic (only after a **new** expense that crosses the limit)

Observed outcomes: All methods returned expected values for valid inputs and rejected invalid inputs gracefully (e.g., blank username, non-digit PIN).

Integration Testing

Integration tests verified interactions among:

- **Activities** (Login, Register, Add/Edit, Main, BudgetSetup)
- **DatabaseHelper** ↔ **SQLite**
- **SessionManager** (login state, budget, currency)
- **FileProvider** for CSV export

Scenarios covered:

- Register → Login → Budget Setup → Dashboard load
- Add/Edit/Delete expense → Dashboard refresh
- CSV export and share intent
- Chart generation from stored data for selected period
- Forgot Password/Username flows using security question

Results: Data flowed correctly across layers; no stale cache issues; CSV exports opened in external apps as expected.

Functional (System) Testing

End-to-end testing was performed against the **approved feature list**:

- Login/Registration
- Forgot Username/Password with security question
- Budget setup (Daily/Weekly/Monthly) & currency
- Expense CRUD + validations
- One-time budget exceed alert (does not repeat on reopen/edit/delete)
- Chart for selected period (consistent with Dashboard period)
- CSV export (proper headers, formatted timestamp, safe note text)

All critical flows worked as designed after fixes.

User Acceptance Testing (UAT)

UAT was conducted by two users (team members) simulating real usage:

- UI clarity and navigation (Login → Budget → Dashboard → Add → Chart)
- Readability of amounts and budget/remaining labels
- Accuracy of chart vs. list totals
- CSV usefulness and readability in spreadsheet apps
- Forgot/Reset flows clarity

Feedback incorporated:

- Enforced **10-digit** mobile number everywhere mobile is entered
- Budget alert shown **only when** limit is crossed by a **new addition** (not on app reopen/edit/delete)
- Eye icon behavior fixed (open = visible, closed = hidden)

Sample Test Cases

Test Case ID	Description	Expected Output	Actual Output	Status
UT-01	SessionManager.set/getBudgetAmount() with double precision	Retrieved amount equals stored value ($\pm$ rounding to 2 dp in UI)	Matched	Pass
UT-02	PIN validation (only digits, length 4–6)	Reject non-digit; accept 4–6 digit PIN	Worked as expected	Pass
UT-03	Mobile validation (exactly 10 digits)	Reject <10 or >10 ; accept exactly 10 digits	Worked as expected	Pass
DB-01	DatabaseHelper.addExpense() inserts row	Row appears with correct amount/category/note/timestamp	Inserted correctly	Pass
DB-02	totalInRange(start,end,userId) sums correctly	Returns accurate period sum	Accurate	Pass
DB-03	categoryTotals() returns map totals	Map keys=categories; values=sum per category	Accurate	Pass
INT-01	Register $\rightarrow$ Login	New user can login immediately	Success	Pass
INT-02	Login via username OR mobile	Both identifiers work with same PIN	Success	Pass
INT-03	Forgot Password via security Q	Correct answer resets PIN	Success	Pass
INT-04	Budget Setup then Dashboard	Dashboard reflects period & currency; labels update	Correct	Pass
INT-05	Add Expense triggers budget check	Alert shows only once when limit is crossed by new addition	As expected	Pass
INT-06	Edit/Delete expense	No budget popup on edit/delete; Dashboard refreshes	As expected	Pass
INT-07	Chart vs. Dashboard period	Chart totals equal period totals used on Dashboard	Consistent	Pass
FUNC-01	CSV export (current period)	CSV has header amount,category,note,timestamp; values formatted; sharable via FileProvider	Correct, opens in Sheets/Excel	Pass
FUNC-02	Theme toggle (if enabled)	App switches between light/dark; persists session theme	Works	Pass
UAT-01	Usability of Login/Register	Clear, easy; error messages guide user	Positive	Pass
UAT-02	Readability with background image	Content readable with overlay; controls tappable	Good	Pass
UAT-03	Eye icon behavior	Open eye $\rightarrow$ PIN visible; Closed eye $\rightarrow$ PIN hidden	Correct	Pass

Results and Output

Overview

The “Expense Tracker” Android Application was successfully developed and executed on an Android environment using Kotlin, XML layouts, and SQLite as the local data management system. The output screens demonstrate the proper functioning of all major modules, including login, authentication, expense management, budget configuration, and data visualization. Each module operates as intended and provides a user-friendly interface that helps users navigate the system effortlessly.

The application begins with a clean and visually appealing **Login Interface**, where users can log in either using a username or registered mobile number. Additional options such as **Register**, **Forgot Password**, and **Forgot Username** assist users in recovering or managing their accounts smoothly. All authentication workflows were tested and performed accurately without any system failures.

Once logged in, users are directed to the **Main Dashboard**, which clearly displays the total spending, remaining budget, and categorized expenses. Smooth navigation is provided via the **Side Navigation Menu**, from where users can access Profile, Export CSV, and Budget Setup options. The **Add Expense Module** allows users to quickly record new transactions with fields such as amount, category, and notes. These entries are stored securely in the SQLite database and displayed instantly in the expense list.

The system also includes an attractive **Category Chart**, which helps users understand their spending pattern visually. This graphical representation dynamically updates based on the user’s recorded transactions and selected time period (Daily, Weekly, or Monthly).

The **Budget Setup Page** enables users to configure their preferred budget limits and currency type. Whenever a user exceeds the defined budget, the system provides a clear alert message, fulfilling the requirement of budget monitoring.

Additional modules such as **Forgot Password**, **Forgot Username**, and **Master Reset** were thoroughly tested to ensure complete data integrity and secure access. All screens maintain a soft color theme and rounded-corner buttons for a simple and modern UI/UX experience.

Overall, the system achieves all the objectives set at the beginning of the project—effectively managing expenses, offering intuitive navigation, ensuring data security, and presenting clear visual insights. The screenshots below display the final working output of the application, validating each component of the project.

Screenshots and Description

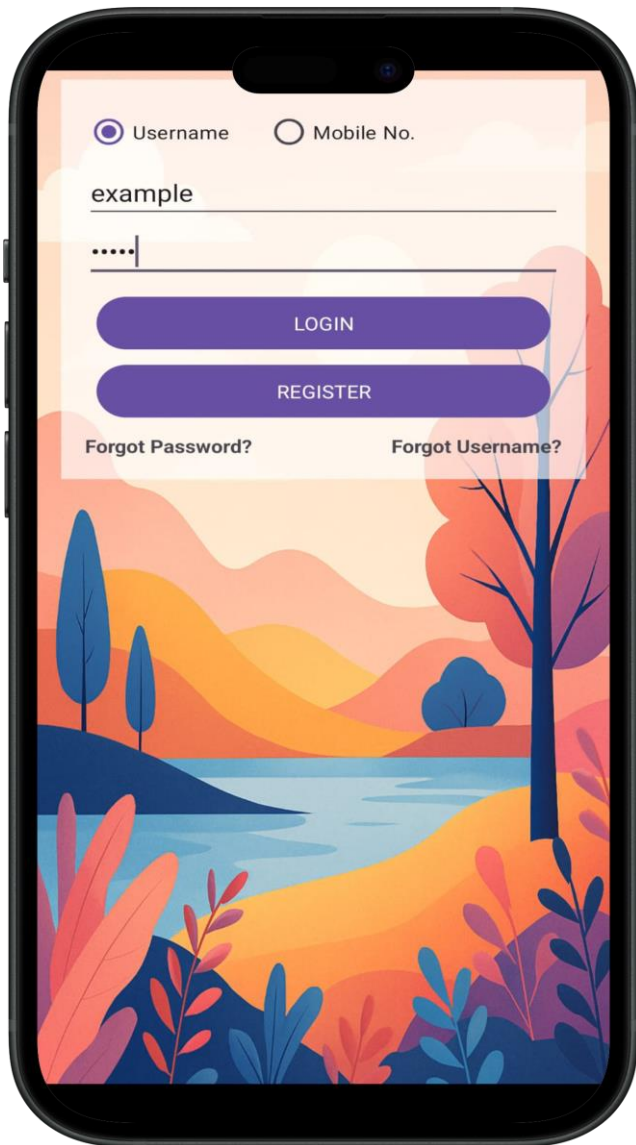


Figure 1 – Login Screen

(Allows the user to log in using either username or mobile number with secure PIN input.)

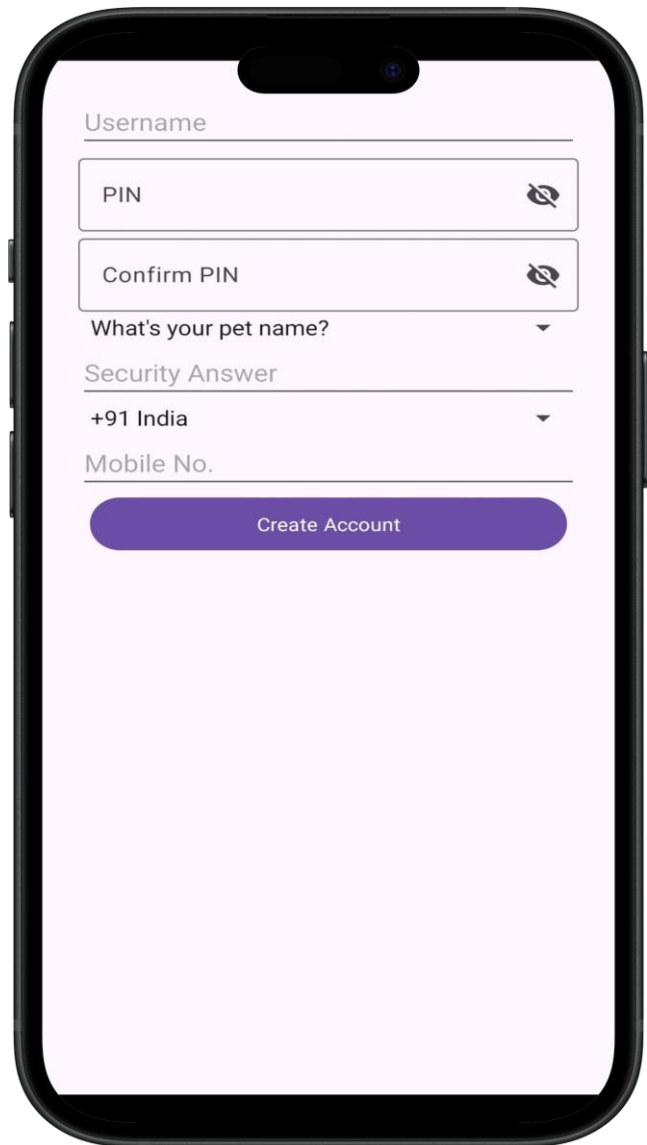


Figure 2 – Register Screen

(Enables new users to create an account with PIN, security question, and mobile number.)

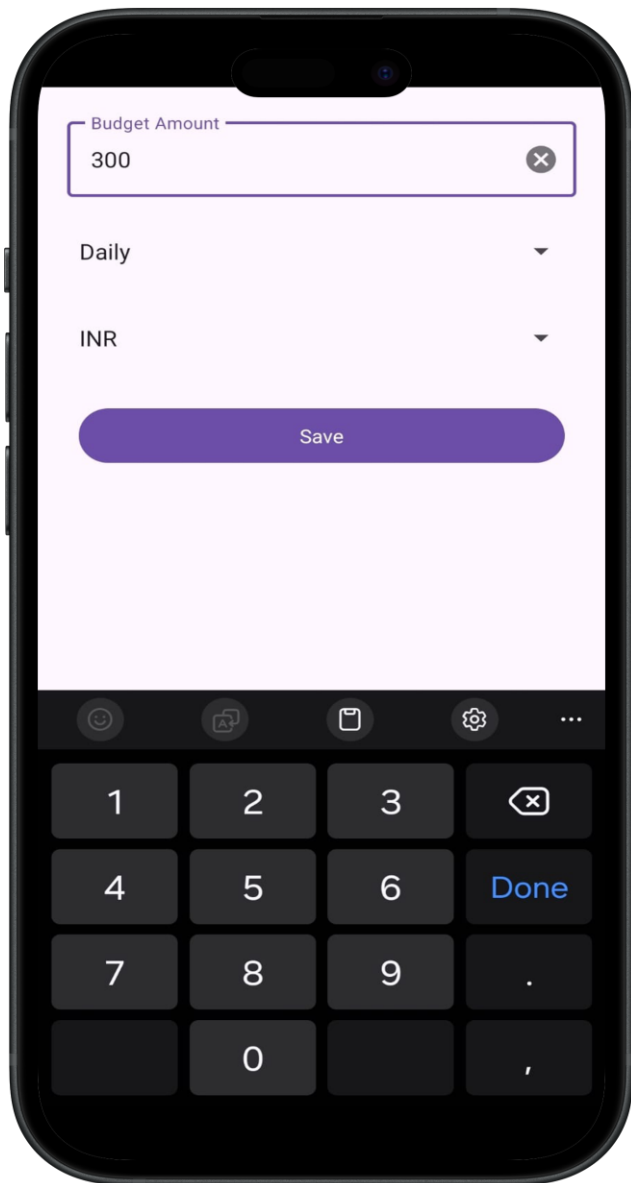


Figure 3 – Budget Setup Screen

(Lets users set a daily/weekly/monthly budget and select preferred currency.)

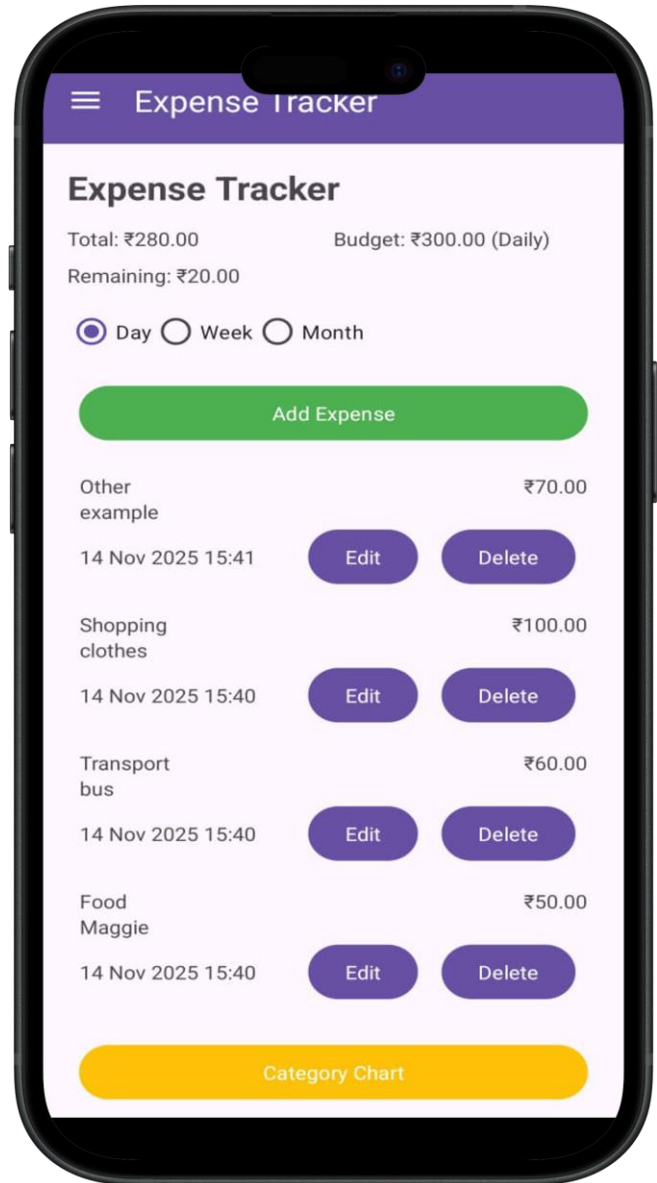


Figure 4 – Dashboard Screen

(Displays total, budget, remaining balance, and recent expense list with edit/delete options.)

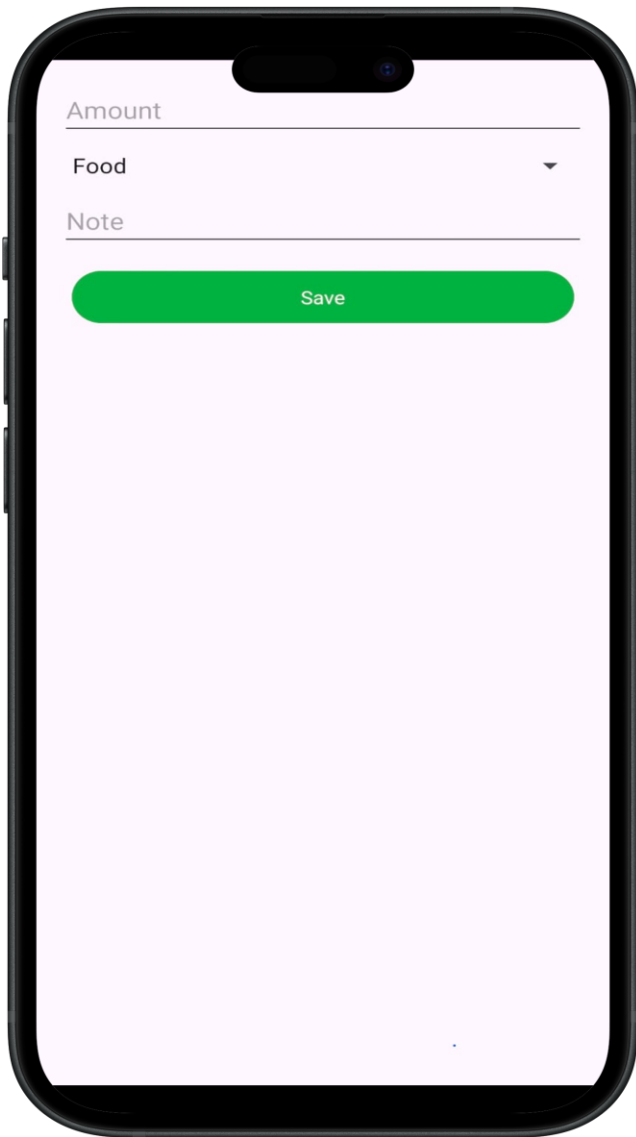


Figure 5 – Add Expense Screen

(Allows users to enter expense amount, choose category, add notes, and save the entry.)

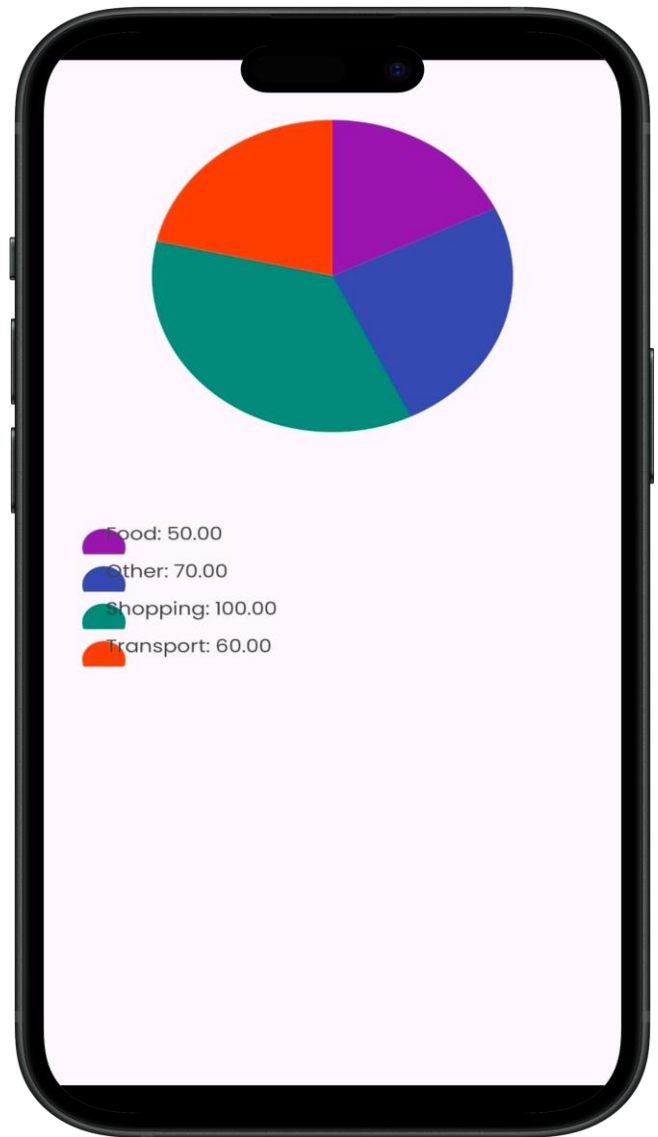


Figure 6 – Category Chart Screen

(Shows a colorful pie chart representing expenses category-wise for the selected period.)

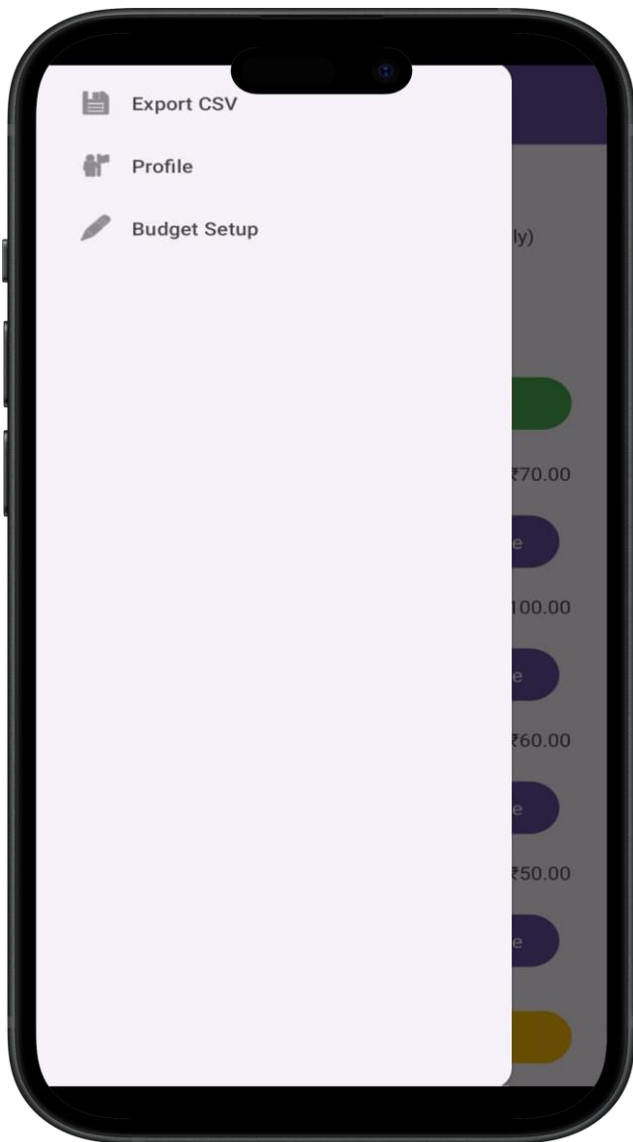


Figure 7 – Navigation Drawer (Menu)

(Provides quick access to Export CSV, Profile, and Budget Setup sections.)



Figure 8 – User Account Management Screen

(Includes Logout and Master Reset options to manage account and clear all data.)

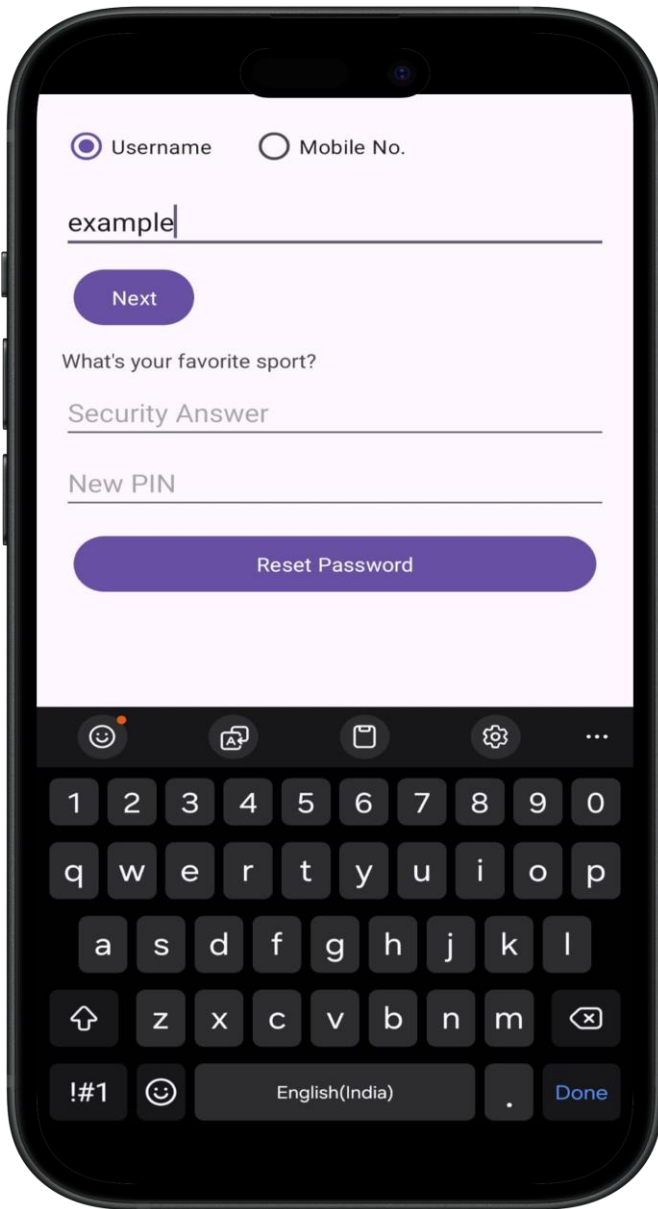


Figure 9 – Forgot Password Screen

(Helps users reset PIN using their registered security question and answer.)

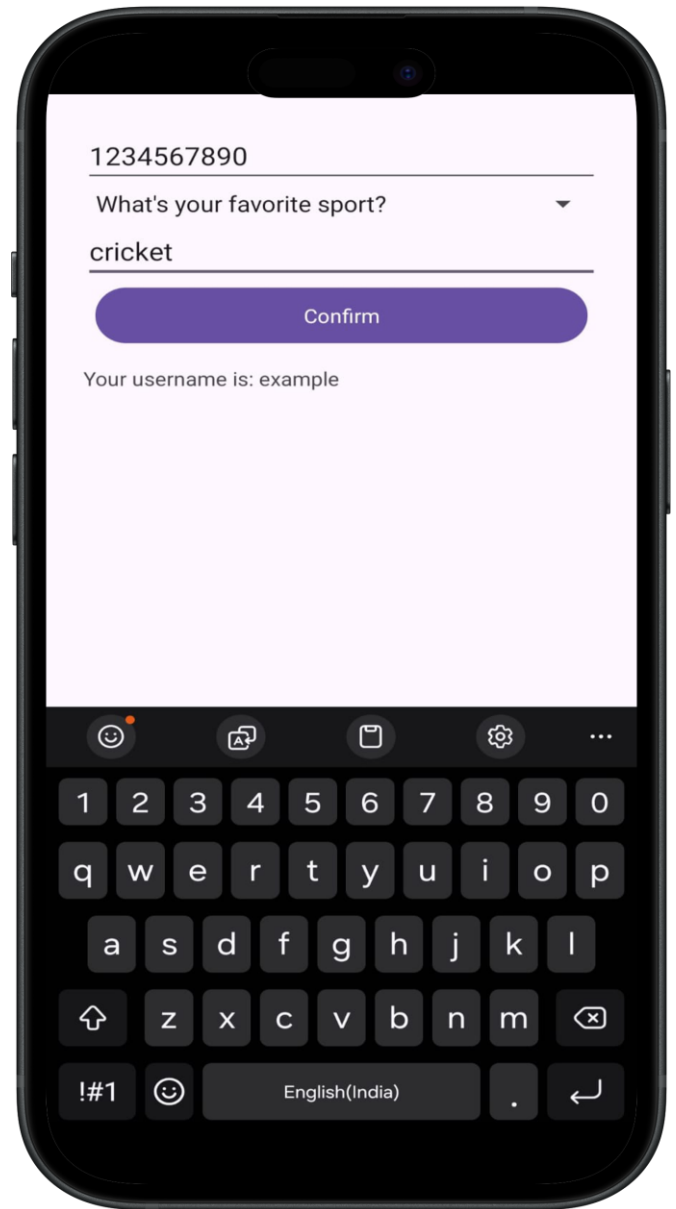


Figure 10 – Forgot Username Screen

(Allows users to retrieve their username using mobile number and security question.)

Conclusion

The “**Expense Tracker Android Application**” marks a significant step toward simplifying personal financial management for everyday users. Through the development of this project, I successfully designed and implemented a system that allows individuals to conveniently record, monitor, and analyze their daily, weekly, and monthly expenses. The application provides a clean interface, intuitive navigation, visually appealing charts, secure login features, and a reliable local database using **SQLite**. By enabling users to add transactions, edit existing entries, set budgets, view category-wise spending summaries, and export their data as CSV files, the application effectively meets its core objective of promoting better financial discipline and awareness. The project demonstrates how a well-designed mobile application can empower users to develop responsible budgeting habits and gain a clearer understanding of their spending behavior.

Along with achieving the functional goals, this project offered immense learning opportunities. The development process strengthened my understanding of mobile application architecture, **Kotlin** programming, **XML-based UI design**, and **database integration**. We learned how to handle user authentication, build responsive layouts, manage activities and navigation, implement custom logic like budget alerts, and generate interactive visualizations through charts. Problem-solving, debugging, and optimizing performance were a major part of the journey, helping us grow both technically and logically. Moreover, this project enhanced our ability to research solutions, resolve errors, manage time, and think from a user-experience perspective — all of which are essential skills for professional software development.

Although the system is fully functional, it still holds great potential for future enhancements. Features such as cloud backup, AI-driven spending insights, automatic receipt scanning, multi-device synchronization, and predictive budgeting can make the application more advanced and intelligent. Additional improvements like push notifications, personalized tips, multi-language support, and theme customization could further enrich the user experience. These possibilities highlight how the project can continue evolving into a more comprehensive personal finance assistant.

In conclusion, the “Expense Tracker Android Application” has successfully achieved its intended goals while providing a strong foundation in mobile app development. It not only enhanced our technical proficiency but also demonstrated the practical application of concepts learned throughout the course. The project stands as a valuable learning experience and serves as a stepping stone toward building more innovative and user-centric applications in the future.

References

Books & Printed Resources

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